

MTH441: Lab 1

In this worksheet, we will NOT use the `lm` function. I would like all of us to be able to learn how to fit a linear regression model without calling in-built functions.

Questions

1. Load the data `table.b1` in package `MPV` which contains data from the National Football League (the wrong football) from 1976.
 - a. Obtain the OLS estimator for the regression coefficients (with intercept).
 - b. Obtain the OLS estimator for σ^2 .
 - c. Verify that the sum of the residuals is 0.
 - d. Verify that $\mathbf{e}^\top \mathbf{X} = 0$.
 - e. Verify that $SS_T = SS_{Res} + SS_R$.
 - f. Obtain the OLS estimator of β for a model without intercept and see if the sum of the residuals is 0 and if $\mathbf{e}^\top \mathbf{X} = 0$.
2. (Montgomery 7.2) A solid fuel rocket propellant loses weight after it is produced. The dataset on Months since production and weight loss (kg) is available in `data(p7.2)` in the R package `MPV`. Fit a simple linear regression model with intercept to fit weight loss (y) versus Months (x). You will need to construct a model matrix yourself.
3. For the above question, run `plot(p7.2)`. What do you observe? Do you believe a “line” is a good choice for a fit here?
4. We will fit a quadratic relationship between Months and weight loss. To the model matrix in 1. add a second column of x^2 and find the OLS estimator of the regression coefficients.
5. On top of the plot in 2., add the linear and the quadratic fits and comment on which one “looks” better.
6. Repeat the following for $k = 3, 10, 50$:
 - a. Generate 1000 draws of $W = \sum_{i=1}^k Z_i^2$ where $Z_i \stackrel{iid}{\sim} N(0, 1)$.
 - b. Plot a histogram of all W s using the `hist` function.
 - c. What happens as k increases?
 - d. Now instead of plotting W s, plot W/k for all values of k . Are the plots now comparable?